ChrisHotelAnalysis

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2026-09-11–2026-09-13

```
#Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code t
# Cleaning data and adding columns for advance time and cancellation in words
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

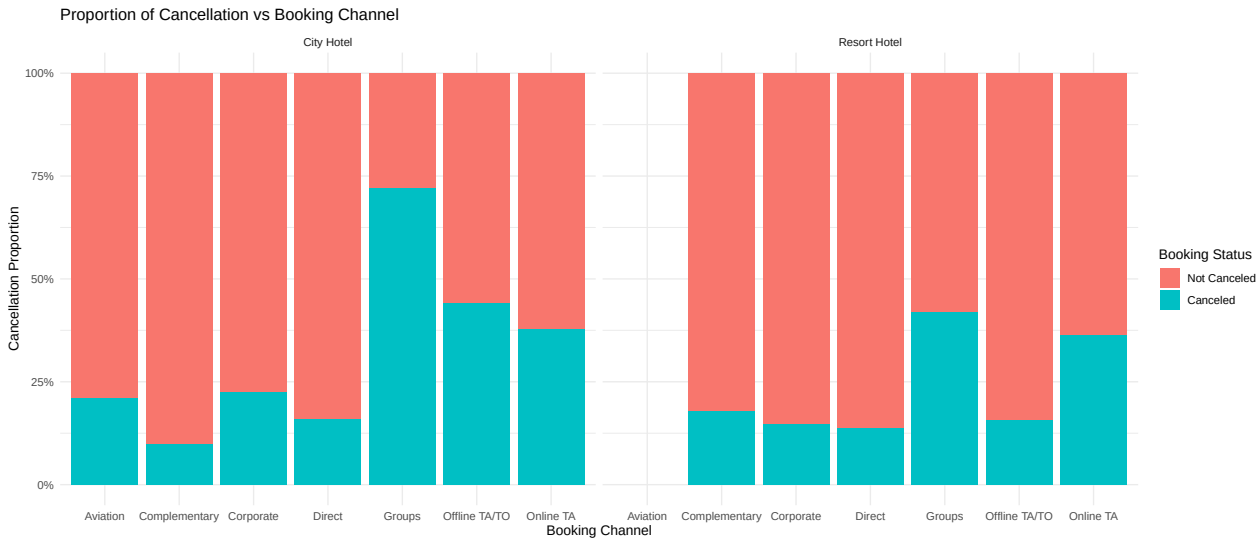
```
library(readr)
df <- read_csv("Downloads/nc_hotel_group.csv")
```

```
## Rows: 20000 Columns: 24
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (8): property, arrival_month, meal_package, booking_channel, reserved_...
## dbl  (15): canceled, advance_time, arrival_year, arrival_week_number, arriva...
## date  (1): status_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
df <- df[df$num_adults != 0 | df$num_children != 0, ]
df <- df[df$weekend_stays != 0 | df$week_stays != 0, ]
df$cancellation_status <- factor(
    df$canceled,
    levels = c(0, 1),
    labels = c("Not Canceled", "Canceled")
)
df <- df %>%
    mutate(
        advance_bucket = cut(
            advance_time,
            breaks = c(-Inf, 7, 30, 90, 180, 365, Inf),
            labels = c(
                "0-7 days",
                "8-30 days",
                "31-90 days",
```

```
                "91-180 days",
                "181-365 days",
                "365+ days"
            )
        )
    )

# bar graph of proportion of cancellations based on channel
ggplot(df, aes(x = booking_channel, fill = cancellation_status)) +
  geom_bar(position = "fill") +
  facet_wrap(~ property) +
  scale_y_continuous(labels = scales::percent) +
  labs(
    y = "Cancellation Proportion",
    x = "Booking Channel",
    title = "Proportion of Cancellation vs Booking Channel",
    fill = "Booking Status"
  ) +
  theme_minimal()
```



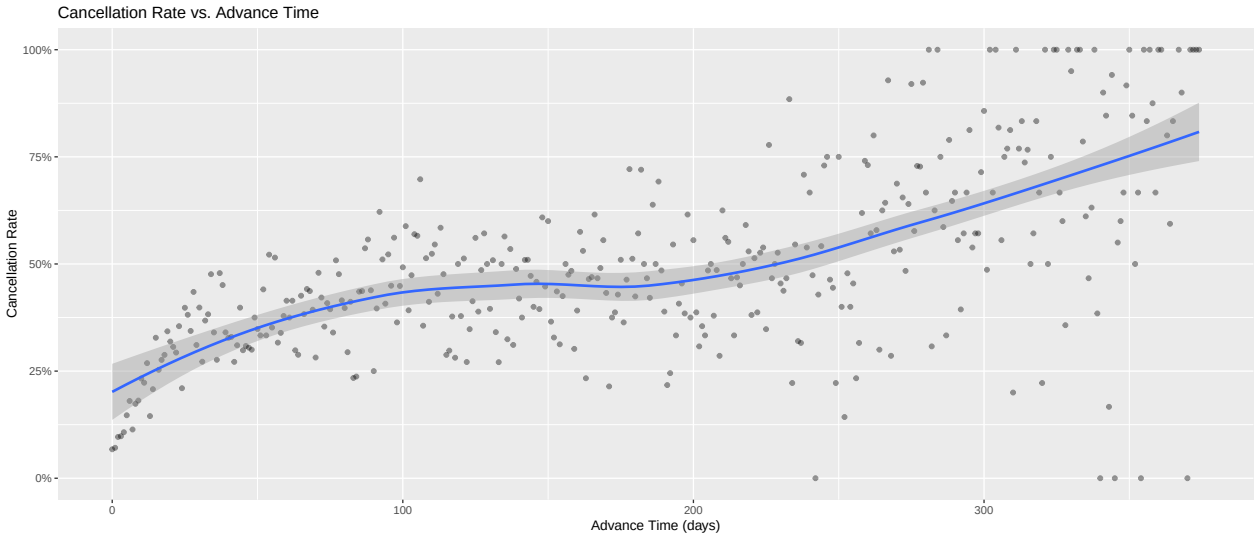

```
# cleaned scatter plot of cancellation rate vs advance time
Q1 <- quantile(df$advance_time, 0.25, na.rm = TRUE)
 Q3 <- quantile(df$advance_time, 0.75, na.rm = TRUE)
 IQR_value <- Q3 - Q1

df_filtered <- df %>%
     filter(
         advance_time >= Q1 - 1.5 * IQR_value,
         advance_time <= Q3 + 1.5 * IQR_value
     )
 cancel_rate_by_advance <- df_filtered %>%
     group_by(advance_time) %>%
     summarise(
         cancel_rate = mean(canceled, na.rm = TRUE)
     )

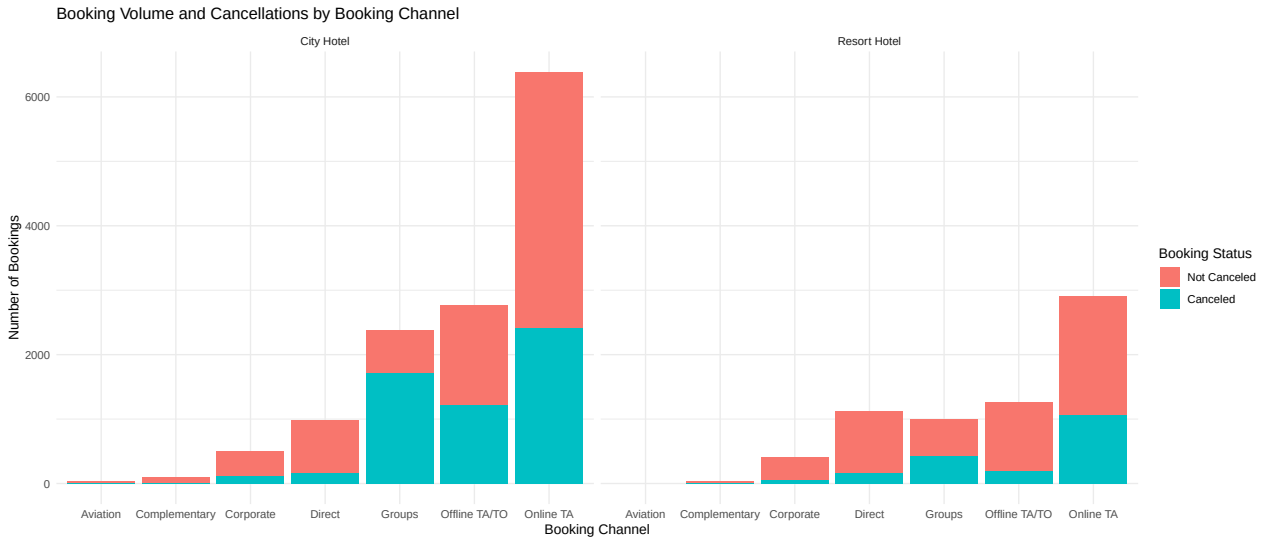
ggplot(cancel_rate_by_advance,
```

```
       aes(x = advance_time, y = cancel_rate)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = TRUE) +
  scale_y_continuous(labels = scales::percent) +
  labs(
      title = "Cancellation Rate vs. Advance Time",
      x = "Advance Time (days)",
      y = "Cancellation Rate"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

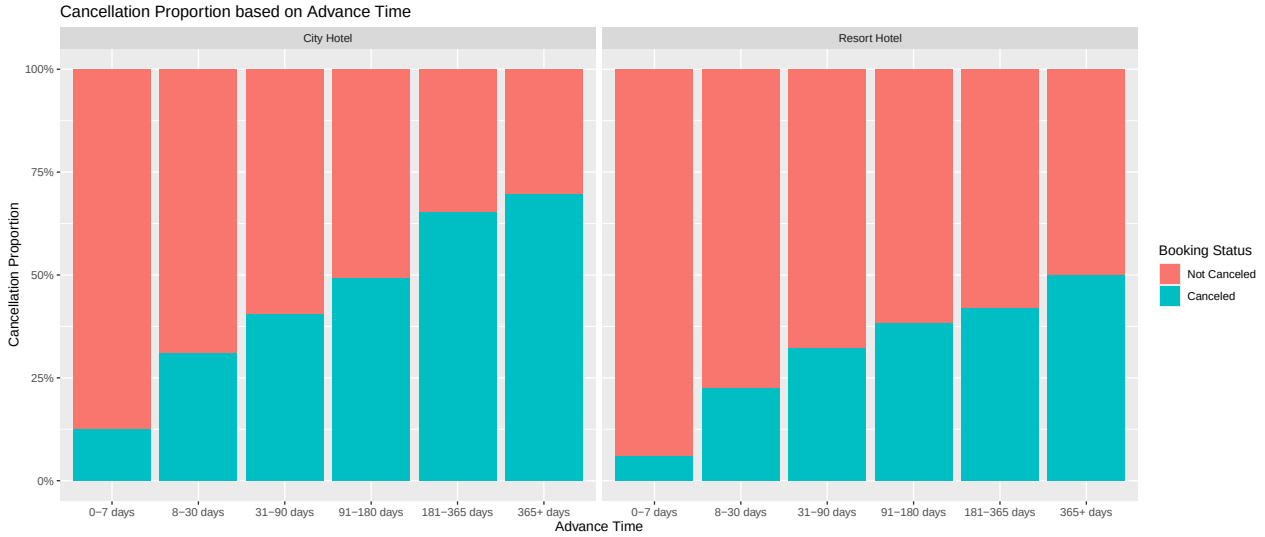



```
# bar graph of volume of cancellations based on channels
ggplot(df,
       aes(x = booking_channel, fill = cancellation_status)) +
  geom_bar() +
  facet_wrap(~ property) +
  labs(
    title = "Booking Volume and Cancellations by Booking Channel",
    x = "Booking Channel",
    y = "Number of Bookings",
    fill = "Booking Status"
  ) +
  theme_minimal()
```




```
# cancellation proportion based on advance time

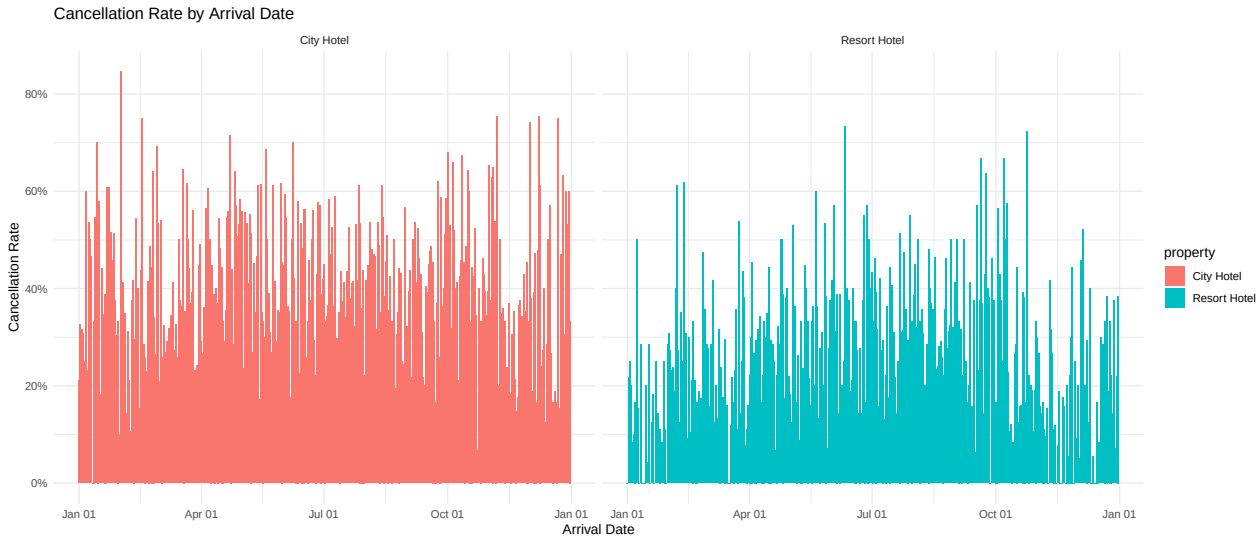
ggplot(df, aes(x = advance_bucket, fill = cancellation_status)) +
    geom_bar(position = "fill") +
    facet_wrap(~ property) +
    scale_y_continuous(labels = scales::percent) +
    labs(
        y = "Cancellation Proportion",
        x = "Advance Time",
        title = "Cancellation Proportion based on Advance Time",
        fill = "Booking Status"
    )
```




```
 #create canellation rate
cancel_rate_by_date <- df %>%
    mutate(
        arrival_date = as.Date(
            paste("2000", arrival_month, arrival_day_of_month),
            format = "%Y %B %d"
        )
```

```
    ) %>%
    group_by(property, arrival_date) %>%
    summarise(
        cancel_rate = mean(canceled, na.rm = TRUE),
        .groups = "drop"
    )

#cancellation rate by arrival date
ggplot(cancel_rate_by_date,
       aes(x = arrival_date, y = cancel_rate, fill = property)) +
    geom_col() +
    facet_wrap(~ property) +
    scale_y_continuous(labels = scales::percent) +
    scale_x_date(date_labels = "%b %d") +
    labs(
        title = "Cancellation Rate by Arrival Date",
        x = "Arrival Date",
        y = "Cancellation Rate"
    ) +
    theme_minimal()
```




```
# total lost revenue
revenue_lost_by_property <- df %>%
  filter(canceled == 1) %>%
  mutate(
    lost_revenue = (weekend_stays + week_stays) * average_daily_rate
  ) %>%
  group_by(property) %>%
  summarise(
    total_revenue_lost = sum(lost_revenue, na.rm = TRUE),
    .groups = "drop"
  )


# total revenue and total potential

revenue_summary <- df %>%
```

```
    mutate(
        revenue = (weekend_stays + week_stays) * average_daily_rate
    ) %>%
    group_by(property) %>%
    summarise(
        total_revenue_earned = sum(revenue[canceled == 0], na.rm = TRUE),
        total_potential_revenue = sum(revenue, na.rm = TRUE),
        .groups = "drop"
    )


# Calculating cancellation_rate
cancellation_rate <- df %>%
    group_by(property) %>%
    summarise(
        total_bookings = n(),
        canceled_bookings = sum(canceled, na.rm = TRUE),
        cancellation_rate = mean(canceled, na.rm = TRUE),
        .groups = "drop"
    )

# average daily rate at each property
average_canceled_rate <- df %>%
  filter(canceled == 1) %>%
  group_by(property) %>%
  summarise(
    average_daily_rate = scales::dollar(
      mean(average_daily_rate, na.rm = TRUE)
    ),
    .groups = "drop"
  )
#Revenue gained from deposits of people that cancelled - assuming that 10% won't even book
potential_deposit_revenue <- df %>%
  filter(
    canceled == 1,
    advance_time >= 200 |
      booking_channel %in% c("Online TA", "Groups", "Offline TA/TO")
  ) %>%
  mutate(
    booking_type = "Require Deposit",
    revenue = (weekend_stays + week_stays) * average_daily_rate,
    deposit_revenue = 0.50 * 0.90 * revenue
  ) %>%
  group_by(property, booking_type) %>%
  summarise(
    potential_deposit_revenue = sum(deposit_revenue, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    potential_deposit_revenue = scales::dollar(potential_deposit_revenue)
  )
```

```r
#Cancellation rate for required group
current_cancellation_rate <- df %>%
  filter(
    advance_time >= 200 |
      booking_channel %in% c("Online TA", "Groups", "Offline TA/TO")
  ) %>%
  group_by(property) %>%
  summarise(
    total_bookings = n(),
    canceled_bookings = sum(canceled == 1, na.rm = TRUE),
    cancellation_rate = mean(canceled, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    cancellation_rate = scales::percent(cancellation_rate)
  )
```